CSE382 | Hotel Booking Machine Learning

Complete Project Report - Tasks 1 to 6
Dataset: City Hotel (H2) Booking Demand
Course: CSE382 Machine Learning
Project repository: AhmedT-7/CSE382-Hotel-Booking-ML

Project at a glance

Workstream	Question	Technique
Tasks 1-2	What is in the dataset, and how can it be prepared safely?	EDA, cleaning, encoding, scaling, splitting
Task 3	What daily room rate (ADR) should we expect?	Supervised regression and tuning
Task 4	Will a reservation be canceled?	Supervised classification and tuning
Task 5	Which booking patterns form distinct groups?	K-Means clustering and profiling
Task 6	Can fewer features preserve useful predictive signal?	PCA, LDA, controlled ML experiments

Executive summary

This project analyzes the City Hotel (H2) booking demand dataset and addresses data understanding, cleaning, supervised ADR prediction, booking cancellation prediction, customer segmentation, and dimensionality reduction. Tasks 1-5 are recorded in separate repository branches; Task 6 is added as a standalone module. Because the branches are not all integrated into `main`, this report synthesizes branch-specific evidence rather than claiming that one combined end-to-end executable already exists.

Key finding: The strongest documented tuned Task 4 classifier is Random Forest (validation ROC-AUC 0.8529). Task 6 substantially compresses the features, although its original-feature baselines retain better validation performance. These results come from different experimental protocols and are not a direct head-to-head contest between Tasks 4 and 6.

Dataset and study scope

The H2 data contains 79,330 rows and 31 columns (Task 1 findings). `ADR` is the regression target and `IsCanceled` the classification target. Approximately 41.73% of bookings were canceled. The study focuses on predictors available at booking creation to avoid information leakage from later outcomes.

The two supervised problems serve different decisions: ADR is continuous, so errors are measured in rate units; cancellation is binary, so discrimination and minority-class detection matter. Clustering has no labeled target and is evaluated with internal cluster metrics and interpretable profiles.

Task 1 - Exploratory Data Analysis and Data Cleaning

Branch: `feature/eda`. The team measured four missing Children values, 24 missing Country values, 8,131 literal NULL Agent values, and 75,641 literal NULL Company values. They identified 25,902 duplicate records and 167 zero-guest records. Outlier counts using a 1.5-IQR rule included 3,387 ADR and 2,576 LeadTime values. EDA charts illustrated right-skewed ADR and lead times, cancellation distribution, and their relationships. Cleaning strips whitespace, fills Children with zero, normalizes selected NULLs, and optionally removes zero-guest records; original rows and outliers are otherwise retained.

Why these decisions matter. Missing-value treatment avoids losing useful records, while retaining potential outliers preserves unusual but valid booking behavior. Keeping identical rows is defensible because two customers may make bookings with identical recorded attributes; later grouped splitting addresses leakage risk from exact duplicates.

```
# Illustrative excerpt of the team's cleaning policy
cleaned_df["Children"] = cleaned_df["Children"].fillna(0)
cleaned_df["Agent"] = cleaned_df["Agent"].replace("NULL", pd.NA)
```

Task 2 - Preprocessing and Feature Engineering

Branch: `feature/preprocessing`. The team built booking-time feature lists for classification and regression, excluded post-booking information, and engineered TotalNights and TotalGuests. Numerical columns receive median imputation and standardization; categorical columns receive constant imputation and one-hot encoding. GroupShuffleSplit divides data into approximately 70% train, 15% validation, and 15% test while keeping hashed duplicate rows grouped. Preprocessing must be fit on training data only.

For example, `TotalNights = StaysInWeekendNights + StaysInWeekNights`, while `TotalGuests = Adults + Children + Babies`. Target leakage is prevented by omitting outcomes (`IsCanceled`, `ReservationStatus`) and attributes that are only known later (such as assigned room type). The preprocessing transformer is fitted on training observations, then applied unchanged to validation and test observations.

```
# Feature engineering illustrated from Task 2
X["TotalNights"] = X["StaysInWeekendNights"] + X["StaysInWeekNights"]
X["TotalGuests"] = X["Adults"] + X["Children"] + X["Babies"]
```

Task 3 - Regression and Optimization

Branch: `Regression`. The regression task predicts ADR. The implementation compares Linear Regression, Decision Tree Regression, and Random Forest Regression. Tree and forest variants undergo GridSearchCV; metrics include MAE, RMSE, and R2. Models and the preprocessing object are serialized with joblib. The inspected branch contains source code, but no committed measured regression score table; numeric performance claims are therefore not made for Task 3.

- **MAE** is the average absolute prediction error and is easy to interpret in ADR units.
- **RMSE** penalizes large pricing errors more heavily.
- **R2** measures improvement over predicting the sample mean (higher is better).

Grid search explores tree depth and minimum split sizes for Decision Trees, and estimator count/depth for Random Forest. A methodological caution: the inspected regression script reports test scores inside its model-selection loop. A stronger final protocol would select on validation or cross-validation, then use the held-out test set only once.

Task 4 - Classification and Optimization

Branch: `feature/classification`. The task predicts booking cancellation with Logistic Regression, Random Forest, and RBF SVM. The baseline report records Logistic Regression validation accuracy 0.7724 and ROC-AUC 0.8199; Random Forest 0.7679 and 0.8311; RBF SVM 0.7748 and 0.8146 (SVM used 5,000 training rows). In the tuning report, Random Forest leads the compared tuned methods: validation accuracy 0.7759, ROC-AUC 0.8529, cancellation F1 0.72. Logistic Regression achieved accuracy 0.7684 and ROC-AUC 0.8202; tuned SVM achieved 0.7591 and 0.8200. Tuning uses validation metrics and grouped cross-validation; final test-set evaluation was not reported.

Tuned model	Accuracy	ROC-AUC	Cancellation F1
Logistic Regression	0.7684	0.8202	0.68
Random Forest	**0.7759**	**0.8529**	**0.72**
RBF SVM	0.7591	0.8200	0.69

Interpretation. ROC-AUC measures how well the model ranks canceled above non-canceled bookings across decision thresholds. Cancellation F1 balances precision and recall for the cancellation class; it is informative when missing cancellations and raising false alarms carry different costs. Random Forest leads on both metrics among the recorded tuned Task 4 models.

The recorded tuning uses three-fold grouped cross-validation, six randomly sampled settings per model, and F1 as the search objective. The best Random Forest setting uses 100 trees, maximum depth 20, minimum leaf size 5, and balanced-subsample class weighting. Its validation confusion matrix is `[[5953, 1036], [1665, 3400]]` (actual classes by rows,

predictions by columns). This corresponds to 3,400 correctly identified cancellations and 1,665 missed cancellations. Threshold choice could still be adjusted based on the cost of these errors.

Task 5 - Clustering and Optimization

Branch: `clustering`. The baseline uses engineered numeric booking attributes and categories, with log-transformation of selected skewed features, scaling, and one-hot encoding. K-Means uses k=4 on 79,163 cleaned bookings, producing 32 encoded features. Its committed baseline metrics show silhouette 0.16406 and Davies-Bouldin 1.68161; an environment-specific fit time of 0.75 seconds is recorded. The branch includes a PCA 2D cluster plot and a cluster profile CSV. These values describe the baseline, not a claim of globally optimal k.

Interpretation. The silhouette score (range -1 to 1) compares within-cluster cohesion against separation from other clusters; 0.164 suggests modest separation in this feature space. Davies-Bouldin is lower when clusters are more compact and separated, but its value is chiefly useful in comparison with alternative configurations. Cluster profiles summarize booking behavior and are more useful for business decisions than raw numeric cluster IDs.

The Member 5 milestone report profiles the groups as follows. ADR and cancellation are used **only to interpret clusters after fitting**, never to form them.

Cluster	Approx. share	Booking pattern	Median ADR	Cancellation rate
0	7%	Family bookings; highest-priced segment	145	35%
1	28%	Last-minute small bookings; median lead time 14 days	96	28%
2	63%	Early planners; more non-refundable deposits	99	49%
3	3%	Frequent corporate guests; almost all repeated guests	66	23%

Cluster 2 is unusually large and may hide smaller subsegments. Member 5's proposed next milestone tests K-Means with k=2-10, sampled agglomerative clustering, and Gaussian Mixture Models using internal metrics, elbow plots, and BIC where appropriate. These are **planned extensions**, not completed comparative results.

Task 6 - Dimensionality Reduction and ML Experiments

Branch: `feature/dimensionality-reduction`. The source file `src/dimensionality_reduction.py` compares original encoded features with reduced features under the same model family for each prediction target. It reads H2, builds booking-time features, uses grouped train/validation/test splitting with random state 42, and fits preprocessing and reduction only on the training subset. **All numeric Task 6 results below are validation results; the held-out test subset was not evaluated.**

6.1 PCA: preserving variance with fewer features

Principal Component Analysis (PCA) constructs new, mutually orthogonal directions that retain as much variance as possible. Here, PCA keeps enough components to explain at least **95% of training-set variance**. It does not know whether a direction helps predict ADR or cancellation, so keeping variance does not necessarily keep every predictive signal.

```
# Excerpt from Task 6: reducer is fitted inside the training pipeline
Pipeline([
    ("preprocessing", preprocessor(X_train)),
    ("reduction", PCA(n_components=0.95, svd_solver="full")),
    ("model", LogisticRegression(max_iter=2000, random_state=42)),
])
```

6.2 LDA: finding separation between cancellation classes

Linear Discriminant Analysis (LDA) uses the `IsCanceled` labels during training to seek directions that separate canceled from non-canceled reservations. For two classes, LDA can produce **at most one** discriminant component (`number of classes - 1`). It is therefore appropriate for the binary classification experiment but not directly for continuous ADR regression.

```
LinearDiscriminantAnalysis(n_components=1, solver="svd")
```

6.3 Cancellation prediction: validation results

The same Logistic Regression estimator is used for all three representations so the reduction method is the principal difference.

Representation	Encoded / retained features	Accuracy	F1	ROC-AUC
Original	58	**0.76357**	**0.62497**	**0.79891**
PCA (95% variance)	21	0.75719	0.61841	0.78855
LDA	1	0.75114	0.62488	0.78885

What changed? PCA removed 37 of 58 encoded dimensions (63.8% reduction), while LDA represented the same input with one discriminant direction (98.3% reduction). The original representation had the highest accuracy and ROC-AUC. LDA's F1 (0.62488) was almost identical to the original model's F1 (0.62497), despite far greater compression. The absolute ROC-AUC difference between original and PCA was about 0.0104.

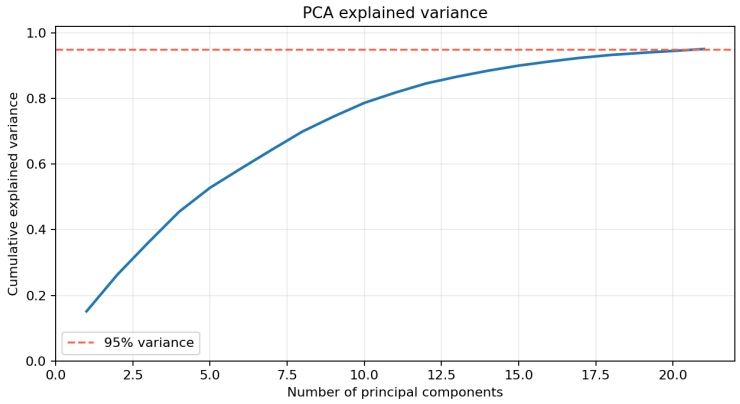


Figure 1: PCA cumulative explained variance for cancellation prediction

6.4 ADR prediction: validation results

The regression comparison uses Ridge Regression for both the original and PCA representations, avoiding a change of estimator between conditions.

Representation	Encoded / retained features	MAE	RMSE	R2
Original	213	**16.96986**	**23.19222**	**0.65124**
PCA (95% variance)	28	20.50503	28.19475	0.48456

PCA reduced the feature dimension by approximately 86.9%, but RMSE increased by about **21.6%**, and R2 fell from 0.651 to 0.485. For this baseline, the variance discarded by PCA includes information useful for ADR prediction, or the rotation and compression make the linear fit less effective. A 95% variance threshold is therefore not automatically optimal for prediction.

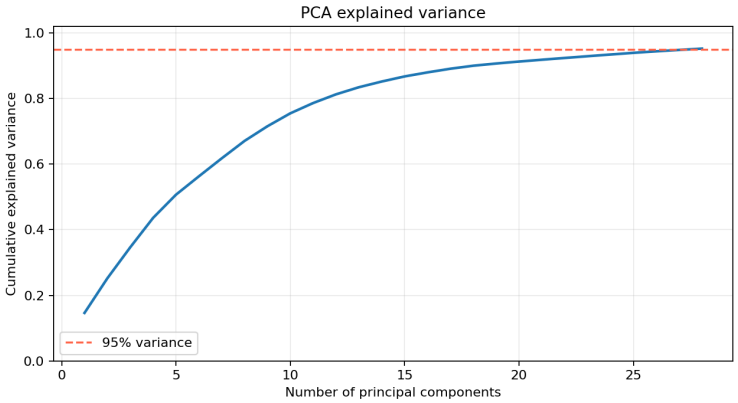


Figure 2: PCA cumulative explained variance for ADR prediction

6.5 Experimental safeguards and limitations

- Each pipeline fits imputation, scaling, one-hot encoding, and PCA/LDA on training data only, preventing validation-set fitting leakage.
- Grouped random splitting keeps identical full records together but does not model changes over time; temporal validation would be useful for deployment.
- Task 6 uses Logistic Regression and Ridge, whereas Tasks 3-4 test additional model families and tuning. Their reported scores are **not directly comparable as identical experiments**.
- Time measurements are hardware-dependent; dimensionality reduction may save prediction/storage cost even if its fitting stage takes additional time.
- Selecting a final model and reporting one test-set evaluation remain future work.

Task 6 conclusion. PCA and LDA demonstrate substantial dimensionality reduction, but under these controlled baselines, the original features provide the best validation predictive quality. The compact alternatives may still be appropriate when memory, speed, or visualization are priorities.

Integration, caveats, and next steps

The branches use related preprocessing ideas but are not yet a fully merged codebase. The regression branch evaluates baseline and tuned models on the test set during its model loop; future comparisons should instead reserve test data until final model selection. Tasks 4 and 6 use validation reporting and keep the test set unexamined. Grouped random splits mitigate identical-row leakage but do not establish temporal generalization. Future integration should unify the feature policy, record environment versions, use one shared split artifact, and perform a single held-out final test comparison.

Overall conclusion

The team has built complementary analytical components: EDA, structured preprocessing, supervised models, clustering, and dimensionality reduction. Tuned Random Forest is currently strongest among the documented Task 4 classifiers. Task 6 demonstrates that aggressive compression trades away validation accuracy for compact representations in these baseline models. These findings motivate further tuning, integration, and rigorous final testing rather than a universal recommendation for or against reduction.

Evidence sources

Repository: https://github.com/AhmedT-7/CSE382-Hotel-Booking-ML . Evidence reviewed: Task 1 EDA findings; Task 4 classification baseline and tuning reports; Member 5 *Clustering: K-Means Baseline* milestone document and the clustering branch's JSON metrics; the `src/` implementations on relevant branches; and the generated Task 6 metrics and figures. Documented Milestone 2 items are plans rather than observed outcomes.